Genetic Algorithm Challenge 2022

Anatomy of the Fittest Algorithm

Genetic Algorithm Challenge 2022

1 Santa’s Workshop Staffing Problem!

6 Genetic Algorithms!

60 second time budget!

Santa’s Workshop Staffing Problem

Problem Overview

There are only 2 weeks left before Xmas and Santa's workshop is buzzing with activity to build all the toys in time to bring Yuletide joy to the 2 billion children worldwide!

Unfortunately for Santa, the **Santa’s Little Workers Union** has demanded some drastic changes to employment conditions to increase fairness and improve elf-esteem:

- No individual elf may work more than 48 hours (i.e. 4 12-hour shifts) in a row
- The most hard-working elf must work a maximum of 50% more hours than the least hard-working elf (in total)
- Junior elf wages must be increased from 2 to 3 cookies per hour
- Senior elf wages must be increased from 3 to 4 cookies per hour

The Work and the Elves

- There are 500,000,000 toys that still need to be made.
- There are 200 elves (150 junior and 50 senior) available to be assigned to the remaining 28 shifts before Xmas Day
- Each senior elf can produce 15,000 toys per hour.
  Each junior elf can produce 10,000 toys per hour.
- There is only space for 150 elves at a time in Santa's workshop during the first week, and 140 elves during the last week (Santa's enormous toy sack displaces 10 elves)

Santa’s Workshop Staffing Problem

Formulation and Representation

Objective Function and Constraints

- **Minimise total cookie costs = 3 (sum of junior hours = J) + 4 (sum of senior hours = S)**
- **subject to:**
    - Toy production requirements (>= 500,000,000)
    - Week 1 workshop capacity constraints (<= 150)
    - Week 2 workshop capacity constraints (<= 140)
    - Maximum consecutive shifts (<= 4)
    - Maximum ratio of maximum to minimum total hours (<= 1.5)
    - 28 x 12 hour shifts

Solution Chromosome

- **# Elves x # Shifts** binary matrix (each 0-1 element is a **gene**)
- Could represent as long binary string, but this is a more natural formulation
  That lends itself to faster computation using matrix algebra (numpy)

Constraint Handling Method

- **Penalty factor method** (add penalty terms to objective function)
- **Penalty weights** derived in challenge notebook to ensure infeasible solution
  Is more expensive than the most expensive known solution

Base Genetic Algorithm

The Base GA uses a 2D chromosome, randomised mutation, single-slice row-wise crossover, and roulette wheel selection

Population Size: 30

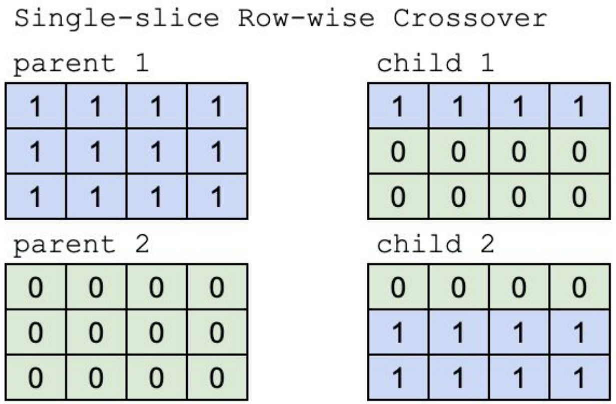
Elite Size: 30

Mutation Rate (per gene): 0.0025

Mutation Mechanism: Randomly flip gene from 0 to 1 and vice versa

Parent Selection Mechanism: Roulette Wheel selection
(selection probability directly proportional to relative fitness)

Crossover Operator: Single-slice row-wise crossover



Untuned GA

Untuned Genetic Algorithm

Selection: Roulette Wheel Selection

Crossover: Single-slice row-wise crossover

Parameters: Population size / Elite size / Mutation rate: **30 / 30 / 0.0025**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **10,491,285**

Variance of fitness over 5 60-second runs: **348,328,159,716**

Average time to minimum: **59.66**

BEST SOLUTION REPORT

Solution is **infeasible**

Solution minimisation objective value: **144,192.0**

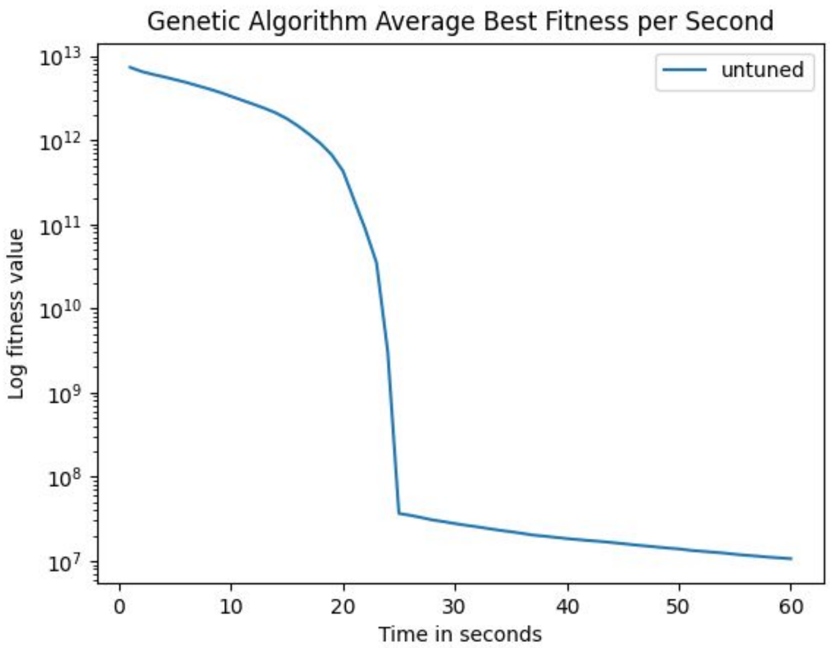
Solution fitness value: **9,557,049**

Toy shortfall violation: **0**

Workshop capacity violation: **0.0**

Consecutive shifts violation: **126.0**

Max/min work ratios violation: **0.071**



Base GA

Base Genetic Algorithm *(manually tuned)*

Selection: Roulette Wheel Selection

Crossover: Single-slice row-wise crossover

Parameters: Population size / Elite size / Mutation rate: **10 / 10 / 0.0025**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **6,480,485**

Variance of fitness over 5 60-second runs: **150,061,370,472**

Average time to minimum: **59.53**

BEST SOLUTION REPORT

Solution is **infeasible**

Solution minimisation objective value: **144,288**

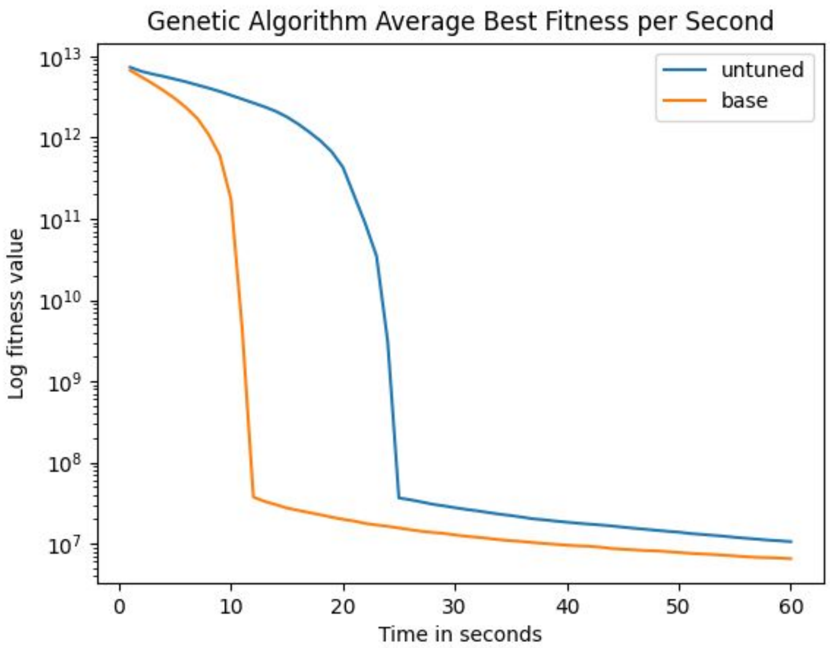
Solution fitness value: **6,155,143**

Toy shortfall violation: **0**

Workshop capacity violation: **2.0**

Consecutive shifts violation: **78.0**

Max/min work ratios violation: **0.071**



Base GA with Numba!

Base Genetic Algorithm + Numba

Selection: Roulette Wheel Selection

Crossover: Single-slice row-wise crossover

Parameters: Population size / Elite size / Mutation rate: **10 / 10 / 0.0025**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **3,110,652**

Variance of fitness over 5 60-second runs: **381,932,576,473**

Average time to minimum: **58.84**

BEST SOLUTION REPORT

Solution is **infeasible**

Solution minimisation objective value: **144,192**

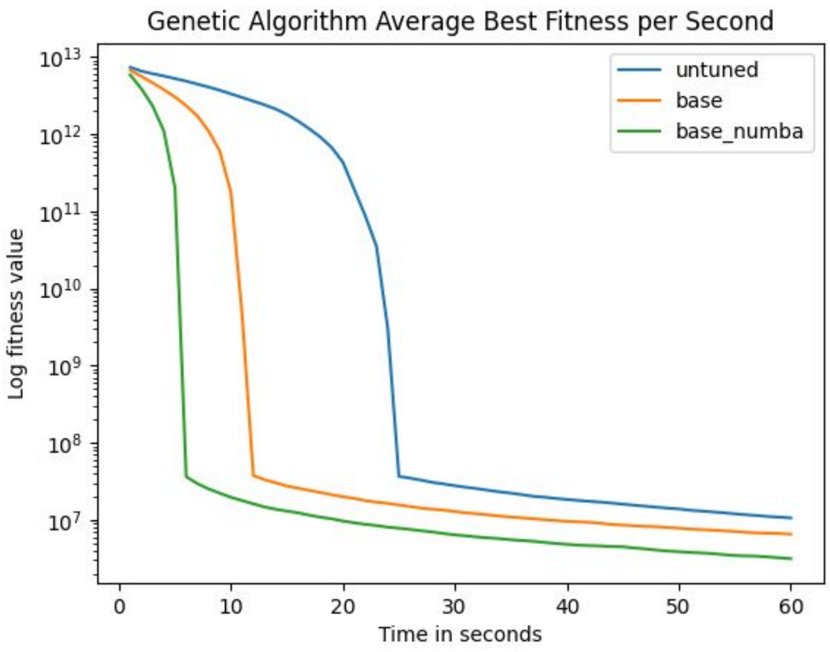
Solution fitness value: **2,436,845**

Toy shortfall violation: **0**

Workshop capacity violation: **0.0**

Consecutive shifts violation: **31.0**

Max/min work ratios violation: **0**



Base GA with Numba and Design Knowledge

Base Genetic Algorithm + Seed the initial population with sensible design/s

Selection: Roulette Wheel Selection

Crossover: Single-slice row-wise crossover

Parameters: Population size / Elite size / Mutation rate: **10 / 10 / 0.0025**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **2,622,850**

Variance of fitness over 5 60-second runs: **879,366,404,687**

Average time to minimum: **56.77**

BEST SOLUTION REPORT

Solution is **infeasible**

Solution minimisation objective value: **144,384**

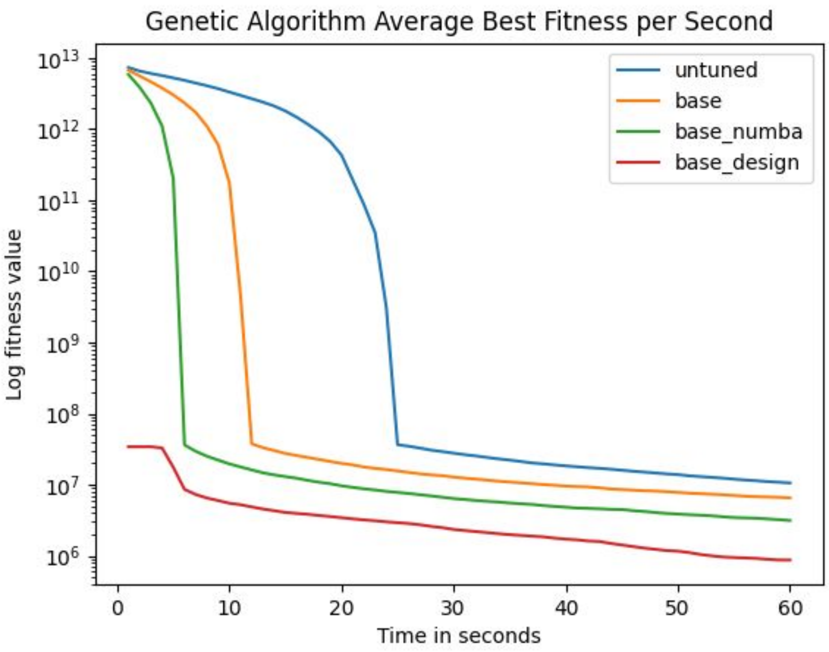
Solution fitness value: **1,179,775**

Toy shortfall violation: **0**

Workshop capacity violation: **0.0**

Consecutive shifts violation: **14.0**

Max/min work ratios violation: **0**



Improved Genetic Algorithm

The Improved GA uses a 2D chromosome, randomised mutation, criss-crossover, and binary tournament selection

Population Size: 32

Elite Size: 29

Mutation Rate (per gene): 0.0025

Mutation Mechanism: Randomly flip gene from 0 to 1 and vice versa

Parent Selection Mechanism: Binary Tournament Selection
(randomly select 2 solutions from population, return fittest)

Crossover Operator: Criss-crossover (see image)

Criss-Crossover

parent 1

1	1	1	1
1	1	1	1
1	1	1	1

parent 2

0	0	0	0
0	0	0	0
0	0	0	0

child 1

1	1	1	0
1	1	1	0
0	0	0	1

child 2

0	0	0	1
0	0	0	1
1	1	1	0

Improved GA with Tournament Selection, Criss-crossover & HPO

Includes Numba

Improved Genetic Algorithm

Selection: Tournament Selection

Crossover: Criss-crossover

Parameters: Population size / Elite size / Mutation rate: **32 / 29 / 0.0025**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **159,076** **(woohoo!)**

Variance of fitness over 5 60-second runs: **878,047,261** **(variance lower, b**

Average time to minimum: **56.89**

BEST SOLUTION REPORT

Solution is **feasible**

Solution minimisation objective value: **144,204** **(not global optimum :(**

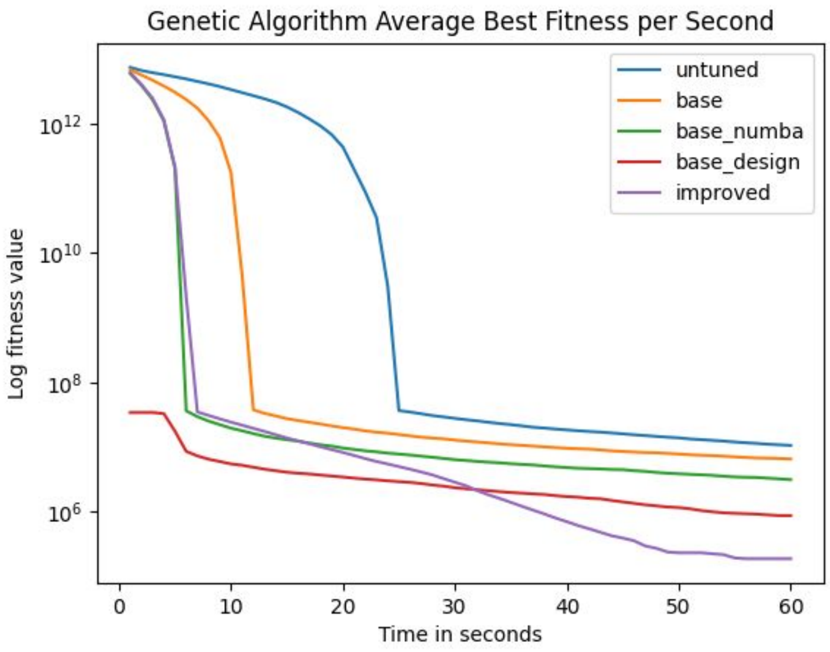
Solution fitness value: **144,204**

Toy shortfall violation: **0**

Workshop capacity violation: **0.0**

Consecutive shifts violation: **0.0**

Max/min work ratios violation: **0**



Improved+++ Genetic Algorithm

The Improved GA uses a 2D chromosome, randomised mutation, criss-crossover, and binary tournament selection

Population Size: 30

Elite Size: 30

Same as the Improved GA plus:

Repair Operator: Randomly Break Consecutive Shifts > 4 with some probability

Random Injection: Create completely random children with some probability

Local Search component: Perform random swaps between elements in the chromosome matrix (with some probability)

Hyperparameter Optimisation: Optimise all the GA parameters using Optuna

Improved+++ GA

Improved with Random Injection, Repair Operator, Local Search and HPO with Optuna

Selection: Tournament Selection

Crossover: Criss-crossover

Parameters: Population size / Elite size / Mutation rate: **30 / 30 / 0.00125**
Random Rate / Repair Rate / Local Search Rate / Swaps: **0.0129 / 0.004 /**

EXPERIMENT RESULTS

Average fitness over 5 60-second runs: **144,216.0**

Variance of fitness over 5 60-second runs: **4,550.4 (super low variance!)**

Average time to minimum: **56.0**

BEST SOLUTION REPORT

Solution is **feasible**

Solution minimisation objective value: **144,144 (global optimal :)**

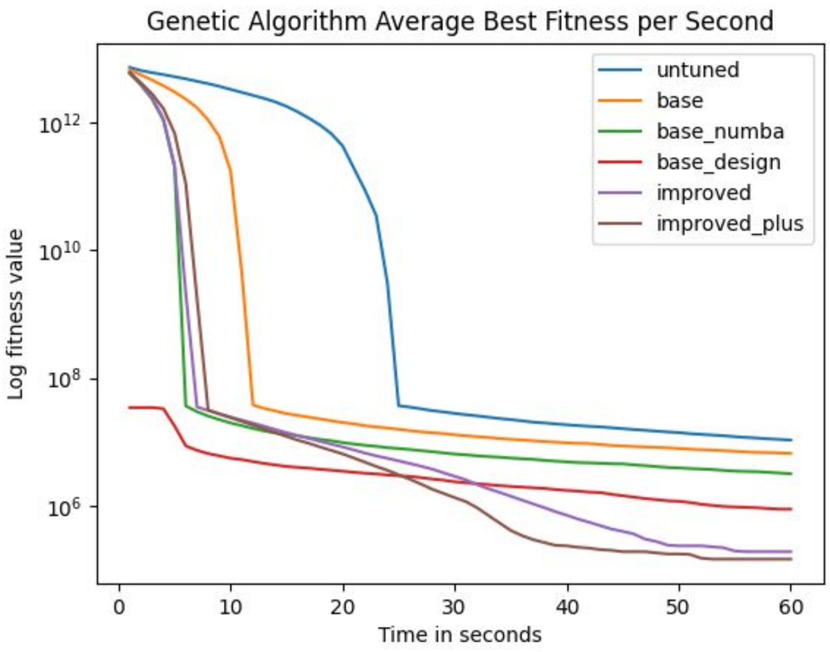
Solution fitness value: **144,144.0**

Toy shortfall violation: **0**

Workshop capacity violation: **0.0**

Consecutive shifts violation: **0.0**

Max/min work ratios violation: **0**



Supporting Code

Code for Santa’s Workshop Problem constants, objective function and constraint violation functions here

Code entry point for optimisation experiment here

Code for improved Genetic Algorithm here

Code for GA parameters here

Code for Optuna HPO here